

IDENTITY-INDEXED LOCAL TEMPO ROUTING FOR MULTI-PERSON REPETITION COUNTING

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ABSTRACT

Multi-person repetition counting estimates a count for each tracked person rather than only a scene total. The task is difficult when people move asynchronously, individual pace changes within a trajectory, and missing poses turn each trajectory into a masked nonstationary signal. We propose an identity-indexed signal-processing specification that receives supplied pose tracks and targets counting without person-wise count or period labels, while acknowledging external supervision in the pose and tracking frontends. Overlapping windows share an encoder and fast, medium, and slow response experts; masked autocorrelation and Fourier evidence yield local period cues and confidence. Soft routing mixes expert responses, and normalized overlap-add reconstructs one response and decodes it once per identity. This design separates per-track state from shared parameters and avoids summing overlapping window counts, which would duplicate events near window boundaries and observed identity gaps. **Experimental evidence is pending synchronization with the collaborator’s frozen implementation and audited MultiRep runs.**

Index Terms— repetition counting, multi-person signals, local tempo, pose trajectories, temporal routing

1. INTRODUCTION

Person-wise repetition counts matter whenever several people exercise, rehabilitate, train, or perform repetitive work in the same scene. A single scene total cannot say who completed each repetition, paused, or changed pace. We therefore study multi-person repetition counting from supplied identity-indexed pose tracks. For N tracks, the primary output is a vector of N counts; their sum is only a diagnostic. The signal is difficult because people move asynchronously, a person’s period may change inside one track, and occlusion creates irregular missing intervals.

Repetition counting progressed from hand-designed periodic motion analysis [1, 2] to learned temporal similarity and correlation models such as RepNet and TransRAC [3, 4]. PoseRAC emphasizes salient poses [5], while PAMS learns period-adaptive multi-scale consistency from pose sequences without count labels in its counting objective [6]. MultiCounter establishes multi-person counting in untrimmed video [7]; MultiCounter+ further addresses identity continuity and long-versus-short repetition regimes [8]. These advances cover multi-person outputs, pose-driven counting, low-label objectives, and broad period adaptation. They do not, however, resolve the narrower signal problem considered here: routing each overlapping window of each supplied identity track according to tempo that changes within that track, then reconstructing one response before count decoding. Recent counters also explore dynamic queries and

decoupled spatiotemporal attention [9, 10]; those architectural directions are complementary to the identity-local temporal scale selection studied here.

The proposed formulation separates inter-person asynchrony from intra-person nonstationarity. Each pose track is a masked multidimensional signal with identity-indexed state, whereas learned encoder and expert parameters are shared across people. The intended counting objective uses neither person-wise count labels nor person-wise period labels, but this qualifier does not extend to the perception stack: pose estimators and trackers may use externally supervised training data. The exact supervision path, including model selection and calibration, remains **SYNC-REQUIRED: supervision firewall** until the collaborator implementation is frozen. Selecting a global long-versus-short regime differs from allowing several pace changes inside one identity trajectory. Mask-aware local estimation also exposes missing observations instead of conflating invalid frames with stationary motion. These distinctions drive both the reconstruction invariant and the planned diagnostic.

Our signal path follows a causal order. A shared encoder represents overlapping masked windows. Local autocorrelation and Fourier evidence estimate period and confidence for each identity-window pair. A soft router mixes fast, medium, and slow shared response experts; normalized overlap-add then reconstructs one continuous response per identity, and one trajectory-level decoder processes it once. Figure 1 shows this specification and its perception boundary. The local worktree does not yet contain the multi-person implementation, so router, expert, reconstruction, and decoder details remain proposal-level interfaces rather than measured components.

The pre-results manuscript makes three falsifiable contributions:

- It formulates person-wise counting as inference on identity-indexed, masked, nonstationary pose signals, with scene totals kept diagnostic.
- It specifies window-local tempo evidence, soft routing over shared response experts, normalized reconstruction, and one decode per track; every source-sensitive interface is explicitly marked for synchronization.
- It preregisters protocol-separated comparisons, paired mechanism ablations, and a piecewise time-warp diagnostic. Empirical claims remain disabled until frozen MultiRep artifacts pass the evidence gates.

2. IDENTITY-INDEXED LOCAL TEMPO ROUTING

The counter begins after perception: it receives pose trajectories with stable identity indices and validity masks. Oracle tracks, tracks from a frozen common frontend, and native method pipelines define different experimental protocols and are never merged into one ranking. Within this boundary, all people share learned parameters, but masks, temporal buffers, periodic evidence, and output responses remain

indexed by identity. Figure 1 summarizes the four-stage proposed specification. Tensor layouts, lifecycle behavior, and all learned modules are **SYNC-REQUIRED: collaborator source binding** in this draft. The separation is structural: no temporal buffer or local period estimate is shared between identities, and shared weights do not imply concatenating people into one temporal signal. This constraint lets the counting stage be audited independently from detection and association errors.

2.1. Masked identity-indexed representation

For person $i \in \{1, \dots, N\}$, let $\mathbf{X}_i$ contain J joints with C coordinates over T_i frames, and let $m_i(t)$ mark a valid observation. Overlapping supports $\mathcal{W}_{i\ell}$ select local windows. A shared encoder $\mathcal{E}_\theta$ maps the masked samples to features, while the primary output remains a person-wise count vector:

$$\begin{aligned} \mathbf{X}_i &\in \mathbb{R}^{T_i \times J \times C}, \quad m_i \in \{0, 1\}^{T_i}, \quad \hat{\mathbf{c}} = [\hat{c}_1, \dots, \hat{c}_N]^\top \\ \mathbf{z}_{i\ell}(t) &= \mathcal{E}_\theta(\mathbf{X}_i[\mathcal{W}_{i\ell}], m_i[\mathcal{W}_{i\ell}])_t, \quad t \in \mathcal{W}_{i\ell} \end{aligned} \quad (1)$$

The shared parameter θ does not mix identity state. Window length and stride, pose normalization, padding, short-track handling, and the creation or retirement of identity buffers are **SYNC-REQUIRED: track tensor and lifecycle contract**. A scene count $\sum_i \hat{c}_i$ may be reported only as a derived diagnostic. The validity mask must accompany every transformation so that padded or occluded frames do not become artificial zero-pose observations. Likewise, overlapping supports are defined on a single identity timeline; a window never borrows samples from another person. These invariants will be tested with permuted identity order, isolated masks, and short tracks once the source interface exists.

2.2. Local masked periodic evidence

Tempo must vary with the window rather than being fixed for the whole track. Let $s_{i\ell}(t)$ denote the scalar evidence signal provisionally projected from $\mathbf{z}_{i\ell}(t)$, and $q_{i\ell}(t)$ its windowed validity mask. We define a mask-normalized non-circular autocorrelation and a normalized Fourier score, then map both to a period estimate and confidence:

$$\begin{aligned} \bar{s}_{i\ell} &= \frac{\sum_t q_{i\ell}(t) s_{i\ell}(t)}{\sum_t q_{i\ell}(t) + \epsilon} \\ A_{i\ell}(\tau) &= \frac{\sum_t q_{i\ell}(t) q_{i\ell}(t + \tau) [s_{i\ell}(t) - \bar{s}_{i\ell}] [s_{i\ell}(t + \tau) - \bar{s}_{i\ell}]}{\sum_t q_{i\ell}(t) q_{i\ell}(t + \tau) + \epsilon} \\ F_{i\ell}(\omega) &= \frac{|\sum_t q_{i\ell}(t) [s_{i\ell}(t) - \bar{s}_{i\ell}] e^{-j\omega t}|^2}{\max_{\omega'} |\sum_t q_{i\ell}(t) [s_{i\ell}(t) - \bar{s}_{i\ell}] e^{-j\omega' t}|^2 + \epsilon} \\ (\hat{p}_{i\ell}, \gamma_{i\ell}) &= \Phi(\{A_{i\ell}(\tau)\}_{\tau \in \mathcal{T}}, \{F_{i\ell}(\omega)\}_{\omega \in \Omega}) \end{aligned} \quad (2)$$

The indices (i, ℓ) make within-track tempo drift explicit. The signal projection, centering rule, lag and frequency domains, temporal units, interpolation, confidence calibration, and low-confidence behavior are **SYNC-REQUIRED: period estimator**. Equation 2 therefore fixes the writing semantics—local masked evidence—without asserting a particular implementation. Autocorrelation exposes repeated similarity at candidate lags, whereas the Fourier score summarizes energy at candidate frequencies. Their joint role is to make the local estimate inspectable rather than to promise that one estimator always dominates. Confidence $\gamma_{i\ell}$ must quantify the quality of available periodic evidence; it cannot be inferred from the selected tempo label after routing. Windows with too little valid support must follow a source-defined fallback or be masked from learning and reconstruction, and the chosen policy must be identical in training and evaluation.

2.3. Soft routing over shared response experts

Let $\mathcal{K} = \{\text{fast}, \text{medium}, \text{slow}\}$ index shared experts. A proposed router consumes the window representation, local period, and confidence, and returns simplex weights. Each expert emits a continuous local response rather than an independently rounded count:

$$\begin{aligned} \mathbf{g}_{i\ell} &= \text{softmax}(\rho_\psi(\mathbf{z}_{i\ell}, \hat{p}_{i\ell}, \gamma_{i\ell}) / \tau_r), \quad g_{i\ell k} \geq 0, \quad \sum_{k \in \mathcal{K}} g_{i\ell k} = 1 \\ \mathbf{r}_{i\ell}^{(k)} &= \mathcal{H}_{\phi_k}^{(k)}(\mathbf{z}_{i\ell}), \quad k \in \mathcal{K} \end{aligned} \quad (3)$$

Soft weights retain uncertainty when a window spans a tempo transition or contains few valid poses. Router inputs, temperature, expert architecture, gradient paths, parameter sharing, and any fallback distribution are **SYNC-REQUIRED: router and expert implementation**. The fast, medium, and slow labels describe the intended scale partition; specialization must later be supported by paired ablations and routing diagnostics. A hard router would discard ambiguity at a transition, while uniform weights would ignore local period evidence. Soft routing retains both as explicit controls: its entropy can be plotted, and hard or uniform behavior can be reproduced without changing the expert outputs. Shared experts also prevent model size from growing with the number of tracked people; this is an architectural specification, not an efficiency result. Any claim about specialization, runtime, or memory remains disabled until the synchronized implementation is profiled.

2.4. Response reconstruction and one track decode

The final stage mixes expert responses before merging overlapping windows. Let $a_{i\ell}(t)$ be a taper on window ℓ , and let $q_{i\ell}(t)$ exclude invalid samples. Normalized overlap-add forms a single track response R_i , after which one decoder extracts events:

$$\begin{aligned} \tilde{r}_{i\ell}(t) &= \sum_{k \in \mathcal{K}} g_{i\ell k} r_{i\ell}^{(k)}(t) \\ R_i(t) &= \frac{\sum_{\ell: t \in \mathcal{W}_{i\ell}} a_{i\ell}(t) q_{i\ell}(t) \tilde{r}_{i\ell}(t)}{\sum_{\ell: t \in \mathcal{W}_{i\ell}} a_{i\ell}(t) q_{i\ell}(t) + \epsilon} \\ \hat{\mathcal{E}}_i &= \mathcal{D}(R_i, m_i), \quad \hat{c}_i = |\hat{\mathcal{E}}_i| \end{aligned} \quad (4)$$

This order prevents duplicated events from independently decoded overlapping windows. Taper choice, denominator behavior, gaps, event confidence, peak parameters, and decoder semantics are **SYNC-REQUIRED: reconstruction and decoder**. Training is likewise unresolved: PAMS-TCC is the documented starting objective [6], while the interface $\mathcal{L}_{\text{PAMS-TCC}} + \lambda_r \mathcal{L}_{\text{route}} + \lambda_t \mathcal{L}_{\text{track}}$ is only a synchronization contract. The latter terms, their existence, formulas, weights, samples, supervision, and gradient paths must be bound to code; any absent term will be deleted. Normalized overlap-add makes boundary handling explicit: each valid frame is divided by the total contributing taper weight before decoding. Invalid gaps remain masked, and one trajectory-level decoder handles events that straddle window boundaries. This invariant is stronger than merely averaging final window counts, because it preserves the continuous evidence on which peak or event decisions depend. A boundary test will place a repetition at several window offsets and verify that the event is decoded once.

The supervision audit covers more than the named loss. It must inspect data loaders, pseudo-target construction, router inputs, validation selection, stopping rules, calibration, and inference-time fallbacks at one commit. Until that firewall passes, the manuscript states only the intended absence of person-wise count and period labels in

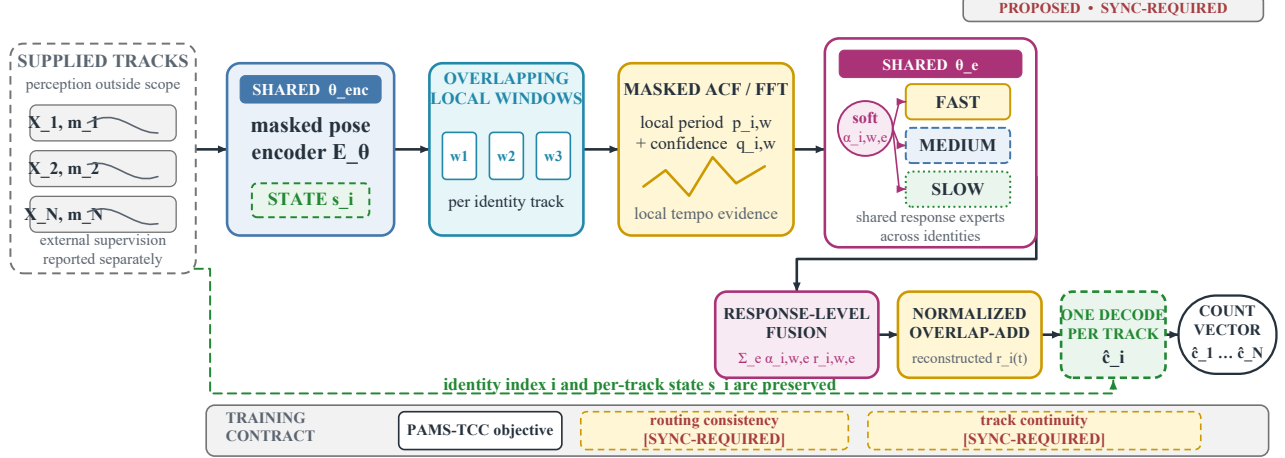


Fig. 1. Proposed signal path, not an implementation or result claim. Supplied pose tracks and masks retain identity-indexed state while sharing encoder and expert parameters. Each window yields local periodic evidence, soft routing mixes continuous expert responses, normalized overlap-add reconstructs one track response, and one decoder returns each person’s count. Red synchronization markers identify interfaces that must be replaced by source-bound definitions.

the counting objective and discloses the external supervision used by pose and tracking frontends.

3. EXPERIMENTS AND ANALYSIS

The experiments are preregistered but not yet executed for the proposed multi-person system. Existing single-person development artifacts are excluded from every table because they neither implement the proposed method nor use the target benchmark. Result prose will be enabled only after a frozen collaborator bundle passes experiment and claim audits.

3.1. Protocol and metrics

The result manifest must bind the MultiRep release, official split, video and annotation checksums, evaluator revision, preprocessing fingerprint, pose modality, supervision path, hardware, seeds, configuration, and raw logs. We separate three protocols: oracle tracks, tracks from one frozen common frontend, and each method’s native pipeline. Native rows provide context but cannot form a matched ranking against common-track rows. Predicted-track evaluations retain HOTA [11], IDF1, and identity switches [12] as frontend diagnostics, separate from the counting metrics. Their proposed-method values remain **PENDING**, **PENDING**, and **PENDING**, respectively, until the same frozen bundle binds them.

The principal counting metrics are Period-mAP, AP50, AP75, AvgMAE, and AvgOBO. Every reproduced learned method requires at least three preregistered seeds; we will report the mean, sample standard deviation, and a paired video-cluster bootstrap 95% confidence interval. A published checkpoint may appear as a contextual row, but it cannot support a variance estimate. Dataset and evaluator fields remain **PENDING** and **PENDING**. The frozen bundle must also contain per-person ground truth, per-person predictions, track identifiers, evaluator outputs, and the exact commands that produced each row. Checksums bind the structured source of every table cell, so the manuscript cannot silently mix releases, frontends, or reruns.

3.2. Main comparison

Table 1 separates native MultiCounter variants from a common-track family. The latter uses one identical tracking frontend for Track-PAMS, a track-global tempo control, and the proposed local router. RepNet and TransRAC [3, 4], PoseRAC [5], dynamic-query counting [9], and dual-branch spatiotemporal counting [10] enter this block only if input, output, tracking, and evaluation parity can be demonstrated. No cell will be emphasized across incompatible protocols. The table will be generated from frozen structured results rather than manually transcribed values. A baseline enters the common-track block only if its temporal input and output can be adapted without privileged labels or method-specific tracking. When such parity cannot be established, the method remains contextual and its unavailable metrics are reported as not applicable rather than imputed.

3.3. Mechanism ablation

Table 2 tests local against global tempo, uniform and hard against soft routing, removal of expert routing, and boundary-reconstruction alternatives. Each retained row must point to one commit and configuration and must reuse paired videos and seeds. Removal rows for routing-consistency, track-continuity, or fallback behavior will appear only if those components exist in the frozen training graph. This rule prevents a writing placeholder from becoming a simulated ablation. Each contrast will report the paired video-cluster confidence interval as well as its seed aggregate, because an unpaired comparison could confound the component change with video difficulty or initialization. The primary ablation metric and direction must be registered before results are inspected.

3.4. Nonstationary-tempo diagnostic

Figure 2 defines the mechanism test. A preregistered piecewise time-warp changes one person’s pace while preserving the evaluation unit and paired identity. The final plot will use either a selection-rule-bound example or an aggregate fixed before inspection. It must record source data, generator hash, warp schedule, routing entropy,

Table 1. Preregistered main comparison. All cells are evidence-controlled; native/contextual and common-track blocks are not ranked across their divider. Arrows specify metric direction, not a performance claim.

Method	Protocol	Period-mAP $\uparrow$	AP50 $\uparrow$	AP75 $\uparrow$	AvgMAE $\downarrow$	AvgOBO $\uparrow$
MultiCounter [7]	native/contextual	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+ [8]	native/contextual	PENDING	PENDING	PENDING	PENDING	PENDING
Track-PAMS [6]	common predicted tracks	PENDING	PENDING	PENDING	PENDING	PENDING
Track-global tempo	common predicted tracks	PENDING	PENDING	PENDING	PENDING	PENDING
Identity-indexed local routing	common predicted tracks	PENDING	PENDING	PENDING	PENDING	PENDING

Table 2. Paired mechanism ablations. Only variants present in the synchronized source remain in the final table; absent conceptual modules are deleted.

Tempo	Router	Boundary fusion	Δ MAE	Δ OBO
global	uniform	normalized OLA	PENDING	PENDING
local	hard	normalized OLA	PENDING	PENDING
local	soft	direct averaging	PENDING	PENDING
local	soft	normalized OLA	PENDING	PENDING

DATA-PENDING DIAGNOSTIC SHELL

local period | routing probability/entropy
reconstructed response | clean-to-corrupt change

Fig. 2. Preregistered nonstationary-tempo diagnostic; the box contains no measured data. A per-person piecewise time-warp will align local period, routing probabilities or entropy, the reconstructed response, and paired clean-to-corrupt changes under a frozen selection rule.

reconstructed response, and clean-to-corrupt metric change. Counts by number of people, occlusion level, and identity-switch severity remain secondary diagnostics. The figure can support a mechanism statement only after the paired artifact and its generation trace pass audit.

The corruption changes tempo inside a track without changing its identity or the number of source events. Clean and warped samples therefore share an evaluation cluster, allowing paired changes to isolate the intended stressor. Occlusion and identity-switch stress tests are generated separately so that a tempo conclusion is not attributed to a simultaneous perception failure. Any representative visualization follows a frozen selection rule; otherwise, the paper reports an aggregate to avoid post hoc example choice.

4. CONCLUSION

We formulated multi-person repetition counting as inference on masked, identity-indexed pose signals whose tempo can change within a track. The proposed specification estimates local periodic evidence, softly routes shared response experts, reconstructs each identity response with normalized overlap-add, and decodes that response once. This pre-results draft has no local multi-person implementation or eligible MultiRep results, and its pose and tracking frontends rely on external supervision. Completion requires a frozen collaborator commit, source-bound losses and interfaces, matched oracle and common-track evaluation, and paired piecewise-tempo diagnostics. Empirical and citation audits must then be rerun before any quantitative or comparative conclusion is enabled.

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licenses, participant provenance, and release terms before submission.

5. REFERENCES

- [1] Ofir Levy and Lior Wolf, “Live repetition counting,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, December 2015, pp. 3020–3028.
- [2] Tom F. H. Runia, Cees G. M. Snoek, and Arnold W. M. Smeulders, “Real-world repetition estimation by div, grad and curl,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018, pp. 9009–9017, IEEE.
- [3] Debidatta Dwibedi, Yusuf Aytar, Jonathan Tompson, Pierre Sermanet, and Andrew Zisserman, “Counting out time: Class agnostic video repetition counting in the wild,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020, pp. 10384–10393.
- [4] Huazhang Hu, Sixun Dong, Yiqun Zhao, Dongze Lian, Zhengxin Li, and Shenghua Gao, “TransRAC: Encoding multi-scale temporal correlation with transformers for repetitive action counting,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, pp. 18991–19000.
- [5] Ziyu Yao and Yuexian Zou, “PoseRAC: Enhancing repetitive action counting with salient poses,” in *Neural Information Processing: 31st International Conference, ICONIP 2024, Proceedings, Part V*, 2025, vol. 15290 of *Lecture Notes in Computer Science*, pp. 255–270, Springer Nature Singapore.
- [6] Shizhao Gao, Jun Li, and Qiming Li, “Count what repeats: Period-adaptive multi-scale consistency for self-supervised repetitive action counting,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*, June 2026, pp. 8143–8152.
- [7] Yin Tang, Wei Luo, Jinrui Zhang, Wei Huang, Ruihai Jing, and Deyu Zhang, “MultiCounter: Multiple action agnostic repetition counting in untrimmed videos,” in *ECAI 2024*, 2024, vol. 392 of *Frontiers in Artificial Intelligence and Applications*, pp. 242–249, IOS Press.
- [8] Yin Tang, Deyu Zhang, Wei Luo, Jinrui Zhang, Wei Huang, Ruihai Jing, and Yaoyue Zhang, “MultiCounter+: Toward efficient and robust multi-instance repetitive action counting,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 36, no. 7, pp. 9462–9476, 2026.
- [9] Xiaoxuan Ma, Zishi Li, Qiuyan Shang, Wentao Zhu, Hai Ci, Yu Qiao, and Yizhou Wang, “Efficient action counting with dynamic queries,” *International Journal of Computer Vision*, vol. 134, no. 8, 2026.
- [10] Xiaoi Wang, Hang Wang, Yan Liu, Huan Hu, and Bruce X.B. Yu, “D²-STX: Decoupling spatial-temporal cross-attention

for dual-branch repetitive action counting,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*, June 2026, pp. 8205–8214.

- [11] Jonathon Luiten, Aljoša Ošep, Patrick Dendorfer, Philip Torr, Andreas Geiger, Laura Leal-Taixé, and Bastian Leibe, “HOTA: A higher order metric for evaluating multi-object tracking,” *International Journal of Computer Vision*, vol. 129, no. 2, pp. 548–578, 2021.
- [12] Ergys Ristani, Francesco Solera, Roger Zou, Rita Cucchiara, and Carlo Tomasi, “Performance measures and a data set for multi-target, multi-camera tracking,” in *Computer Vision – ECCV 2016 Workshops*, Gang Hua and Hervé Jégou, Eds., Cham, 2016, vol. 9914 of *Lecture Notes in Computer Science*, pp. 17–35, Springer International Publishing.